

EECS 151 FPGA Project

ILA Introduction

This page will guide you through defining and instantiating an Integrated Logic Analyzer (ILA) for your design. They give you access to signals in your design even after programming the FPGA with your design, which can help catch any bugs that weren't caught in simulation.

ILA Setup

The following steps will show how to set up an ILA for the PC signal in your design. Once you have done so, feel free to use ILAs to track other signals in your design!

Project Setup

- 1 Open Vivado (`vivado`)
- 2 Under the Quick Start menu, create a new project under your main project repo.
- 3 When asked for project type, you can leave the defaults (RTL project, both boxes unchecked).
- 4 On the Add Sources page, add the `hardware/src` directory. You can keep all other defaults.
- 5 You can skip past the Add Constraints page.
- 6 Select `xczu3eg-sfvc784-2-e` as the part and finish creating the project.

ILA Generation

- 1 Click IP Catalog along the left sidebar. This should focus the IP Catalog tab on the right side of Vivado if it was not already focused.
- 2 In the search bar, type "ILA". We want the ILA (Integrated Logic Analyzer), not the System ILA. Double click it.
- 3 On the next menu, in the General Options, change the Sample Data Depth to 4096. Under the Probe _Ports tab, change the probe width to 32 bits. Click OK.
- 4 Click Generate, leaving the default settings.

Instantiation

- 1 Initialize the ILA in your `cpu.sv` file. There is a module definition in `project_1/project_1.gen/sources_1/ip/ila_0/ila_0_stub.v` (assuming you used the default project name). For now, connect your PC to the probe. Since we only want to generate the ILA on the FPGA, enclose your module definition like the following:

```
`ifdef SYNTHESIS  
// Module instantiation  
  
`endif
```

- 1 In `hardware/scripts/synth.tcl`, add your ILA at the TODO. You should add the following block:

```
read_ip /PATH/TO/ila_0.xci  
generate_target all [get_ips]  
synth_ip [get_ips]
```

Please replace the `/PATH/TO` with the actual path to your ILA's `.xci` file. Most likely this will be in `project_1/project_1.srscs/sources_1/ip/ila_0`.

On the FPGA

- 1 Use the standard `make` flow to program the FPGA with your design.
- 2 In Vivado, go to Tools on the top bar and select Open Hardware Manager.
- 3 Again under Tools, select Auto Connect.
- 4 In Vivado, under the Hardware window (top left), right click `xczu3_0` and select Hardware Device Properties.
- 5 Click the ... next to Probes file and select `build/impl/z1top.ltx`. Now, click the trigger immediate (» icon) button, which will give a waveform of the signal.

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